PHYC10009 foundations of Physics – lecture 9

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Physics, Room 501

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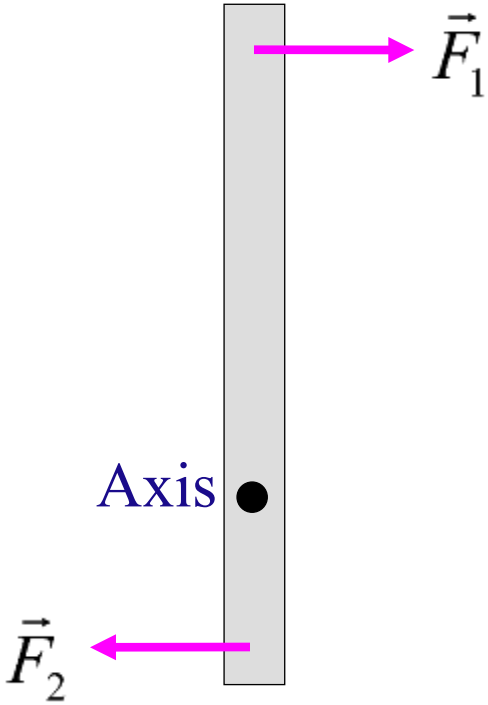
Static equilibrium - G 9.2



- Suppose that in the diagram on the left, we have $\vec{F}_1 = \vec{F}_2$

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 = 0 = m\vec{a}_{com}$$

- The net *force* on the object is zero and the object's centre of mass does not accelerate.
- But, the net *torque* on the object is *not* zero – the object has an angular acceleration (it rotates clockwise).

$$\vec{\tau}_{net} = I\vec{\alpha}$$

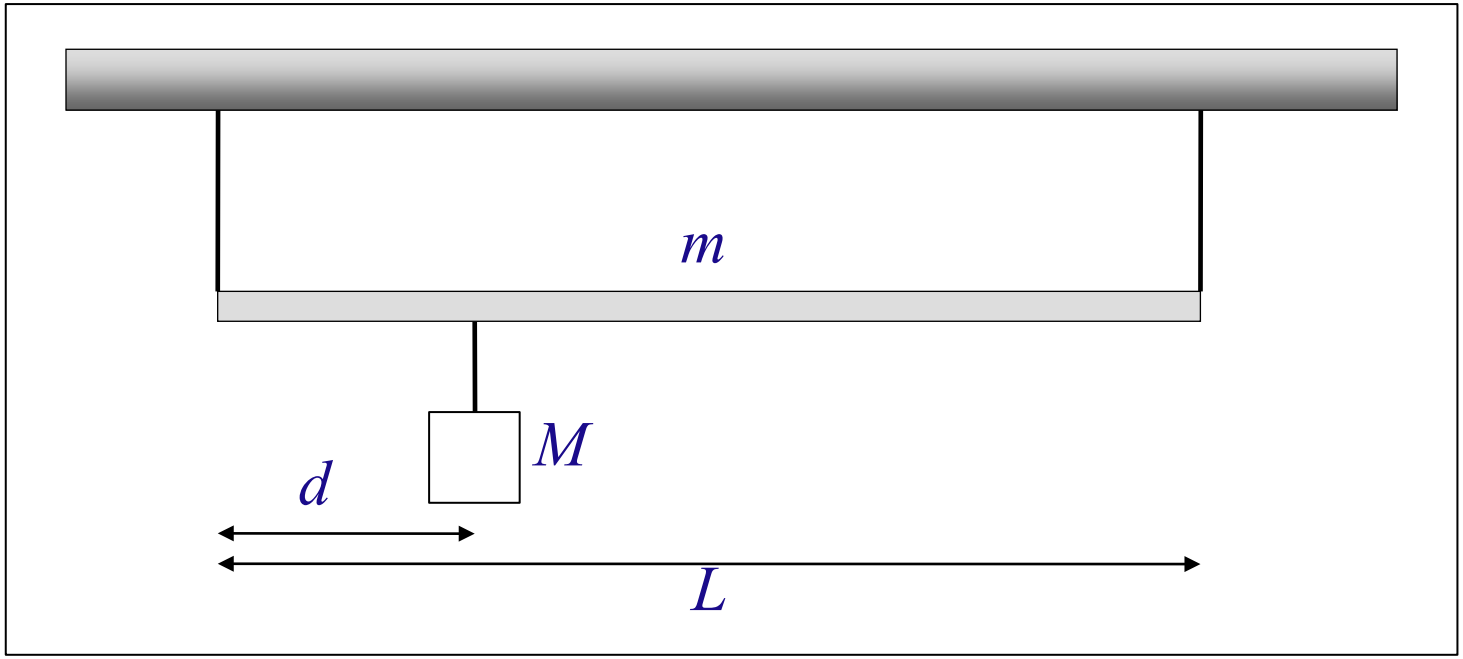
If an object is in static equilibrium:

- Net force is zero.
- Net torque about *any* axis is zero.

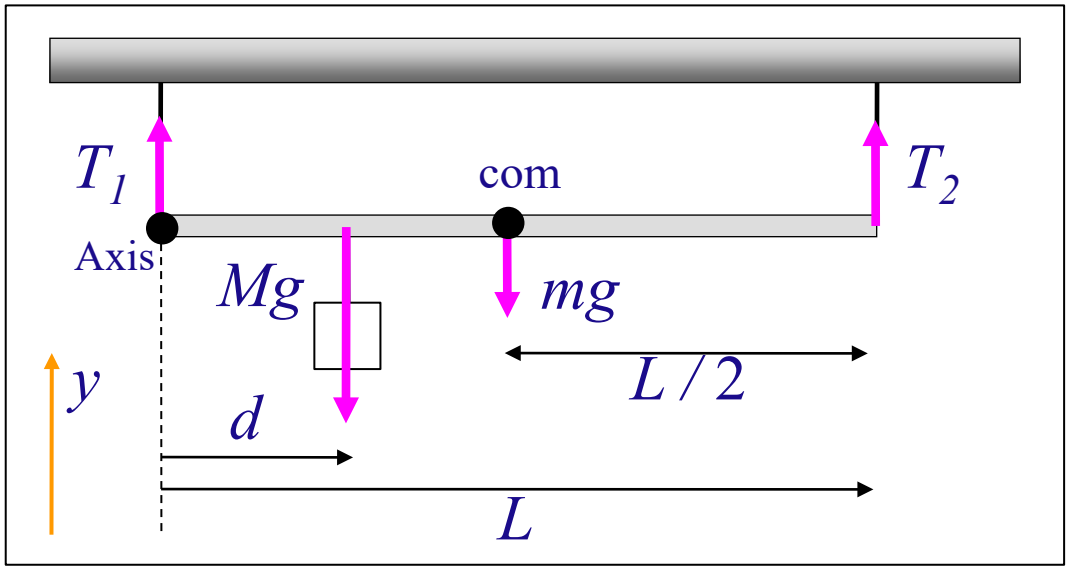
(MI -1 demo)

Milk carton mobile – G 9.2

A rod of mass m = 0.2 kg and length L = 1.0 m is suspended from the ceiling by strings tied to each end, as shown. If a milk carton of mass M = 1.0 kg is hung a distance d = 30 cm from one end of the rod, what are the tensions in each of the supporting strings?



Milk carton mobile – G 9.2



$m = 0.2$ kg, $L = 1.0$ m,
$M = 1.0$ kg, $d = 0.30$ m

$$\tau = rF\sin\phi$$

(We can choose *any* point as the axis.)

Torques around left end of rod

(Note the signs of the torques in this equation.)

$$-Mgd - mg\frac{L}{2} + T_2 L = 0 \qquad T_2 = \frac{Mgd}{L} + \frac{mg}{2} = 3.9\ \text{N}$$

Forces in y direction:

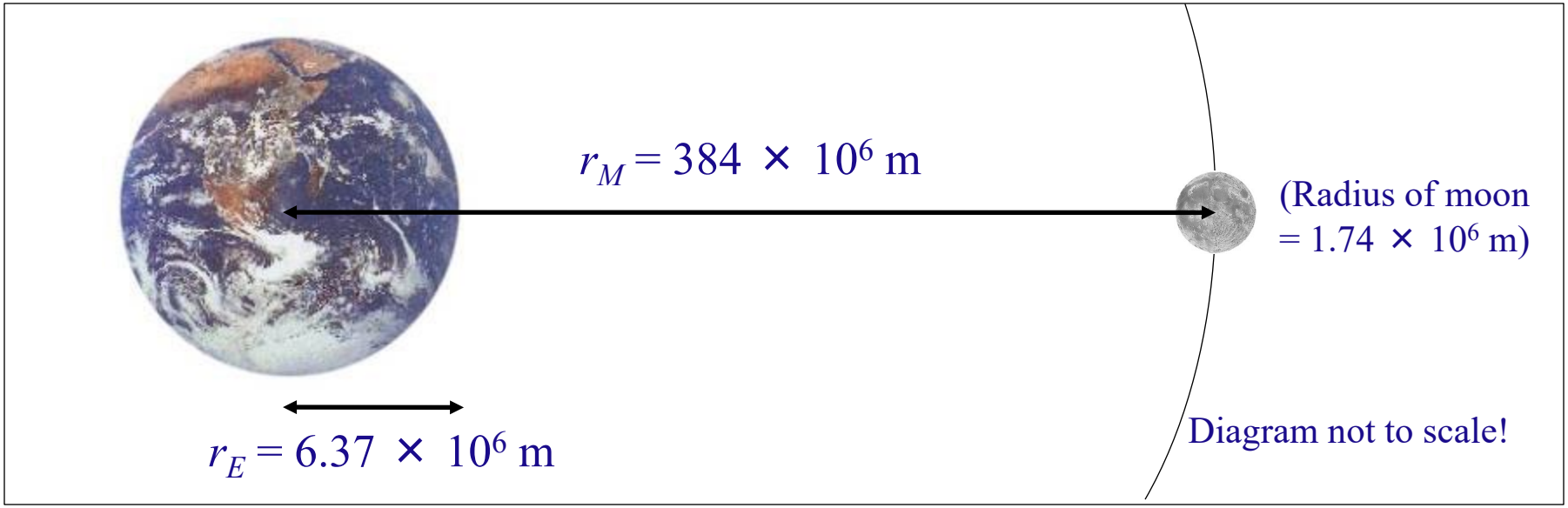
$$T_1 - Mg - mg + T_2 = 0 \qquad T_1 = Mg + mg - T_2 = 7.8\ \text{N}$$

D Moments demo

Isaac Newton and gravity

- "I deduced that the forces which keep the planets in their orbs must be **reciprocally** as the **squares of their distances** from the centres about which they revolve; and thereby compared the force requisite to keep the Moon in her orb with the force of gravity at the surface of the Earth; and found them answer pretty nearly."
  - Newton, *Philosophiae Naturalis Principia Mathematica* (1687).

Checking Newton… - G 5.5



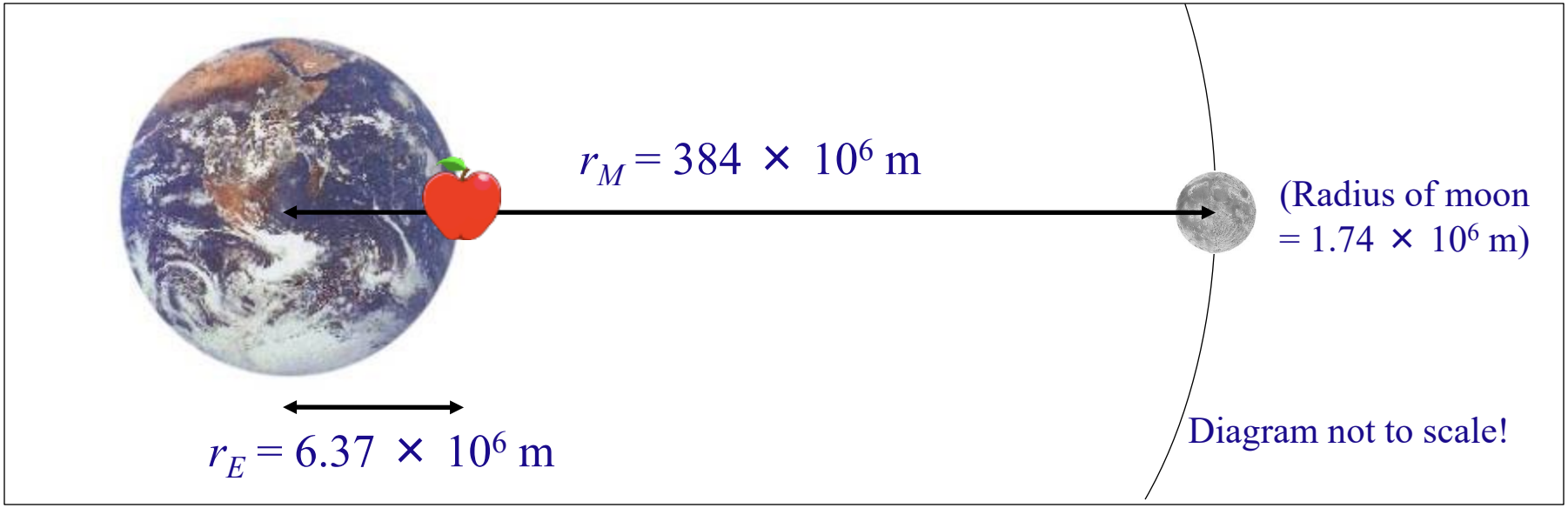
Assume Moon's orbit is a circle of radius r_M.

Period of revolution around Earth: $T = 27.3\text{ days} \times \frac{24\text{ hour}}{1\text{ day}} \times \frac{60\text{ min}}{1\text{ hour}} \times \frac{60\text{ s}}{1\text{ min}} = 2.36 \times 10^6\text{ s}$

Orbital velocity: $\text{v} = \frac{2\pi r_M}{T} = 1023\text{ m/s}$ ~ 3x speed of sound in air

Centripetal acceleration: $a_M = \frac{\text{v}^2}{r_M} = \frac{4\pi^2 r_M}{T^2} = 2.73 \times 10^{-3}\text{ m/s}^2$

Checking Newton… - G 5.5



Assume Moon's acceleration is due to Earth's gravity.

Acceleration of apple due to gravity: $a_E = 9.8 \text{ m/s}^2$

Acceleration of Moon due to gravity: $a_M = 2.73 \times 10^{-3} \text{ m/s}^2$

Newton observed:

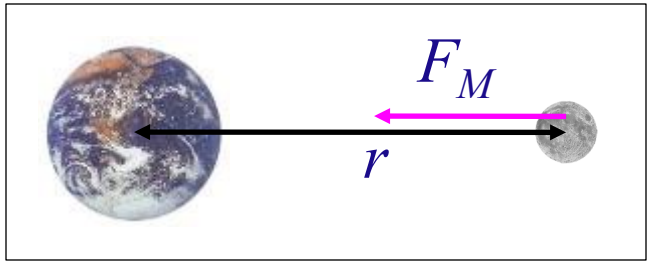
$$\frac{r_E}{r_M} \approx \frac{1}{60} \qquad \frac{a_E}{a_M} \approx 3600$$

So $$\frac{a_E}{a_M} \approx \left(\frac{r_M}{r_E}\right)^2$$

Newton proposed:

$$a_E \propto \frac{1}{r_E^{\,2}} \qquad a_M \propto \frac{1}{r_M^{\,2}}$$

Newton's law of gravity – G 5.5



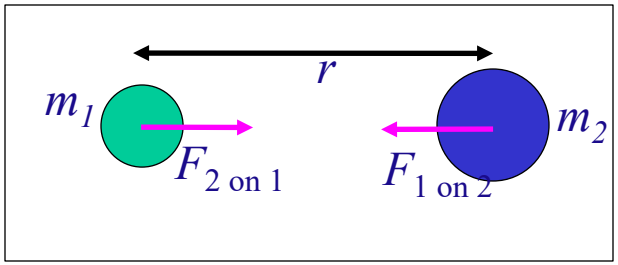
From $a_M = \dfrac{F_M}{m_M}$ and $a_M \propto \dfrac{1}{r^2}$

Newton deduced $\dfrac{F_M}{m_M} \propto \dfrac{1}{r^2}$ $\quad F_M \propto \dfrac{m_M}{r^2}$

Constant of proportionality (determined by experiment – Cavendish 1798)

Newton reasoned $F_M \propto m_E$ $\quad F_M = \dfrac{G m_E m_M}{r^2}$

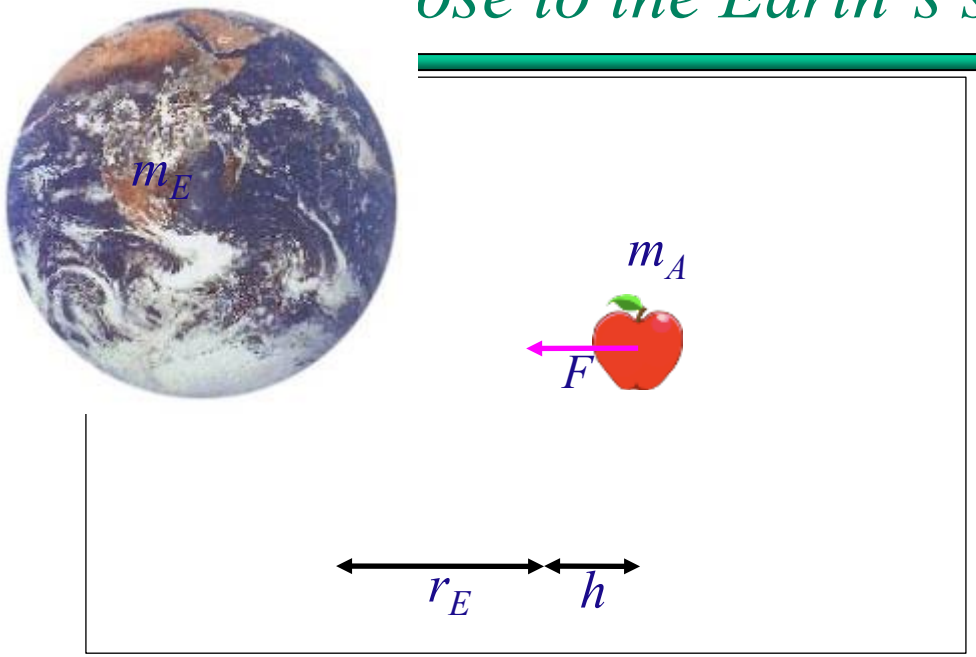
Newton asserted that this law is a *universal* law of physics. It applies to *any* two masses anywhere in the universe.



$$F_{1\text{ on }2} = F_{2\text{ on }1} = \frac{G m_1 m_2}{r^2}$$

Universal gravitational constant: $G = 6.67 \times 10^{-11}\ \mathrm{Nm^2kg^{-2}}$

Gravity close to the Earth's surface – G 5.6



$$F = \frac{G m_E m_A}{(r_E + h)^2}$$

If $h << r_E$

$$F \approx \frac{G m_E m_A}{r_E^{\,2}}$$

But $F = w = m_A g$

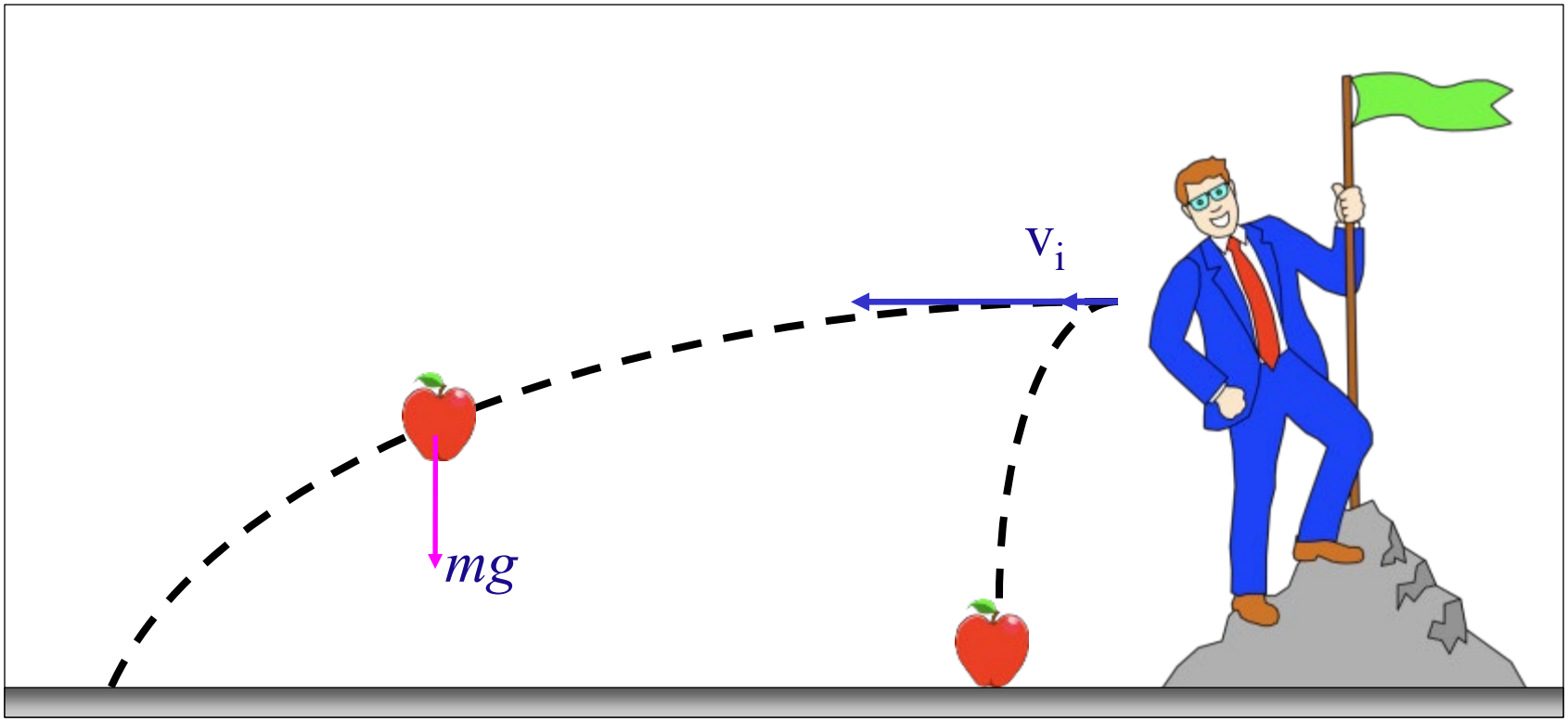
Acceleration of apple due to gravity:

$$g = \frac{F}{m_A} = \frac{G m_E}{r_E^{\,2}} = \frac{(6.67\times10^{-11})(5.98\times10^{24})}{(6.37\times10^{6})^2} = 9.83\ \mathrm{m/s^2}$$

Near the Earth's surface, acceleration due to gravity is $g = \dfrac{G m_E}{r_E^{\,2}}$

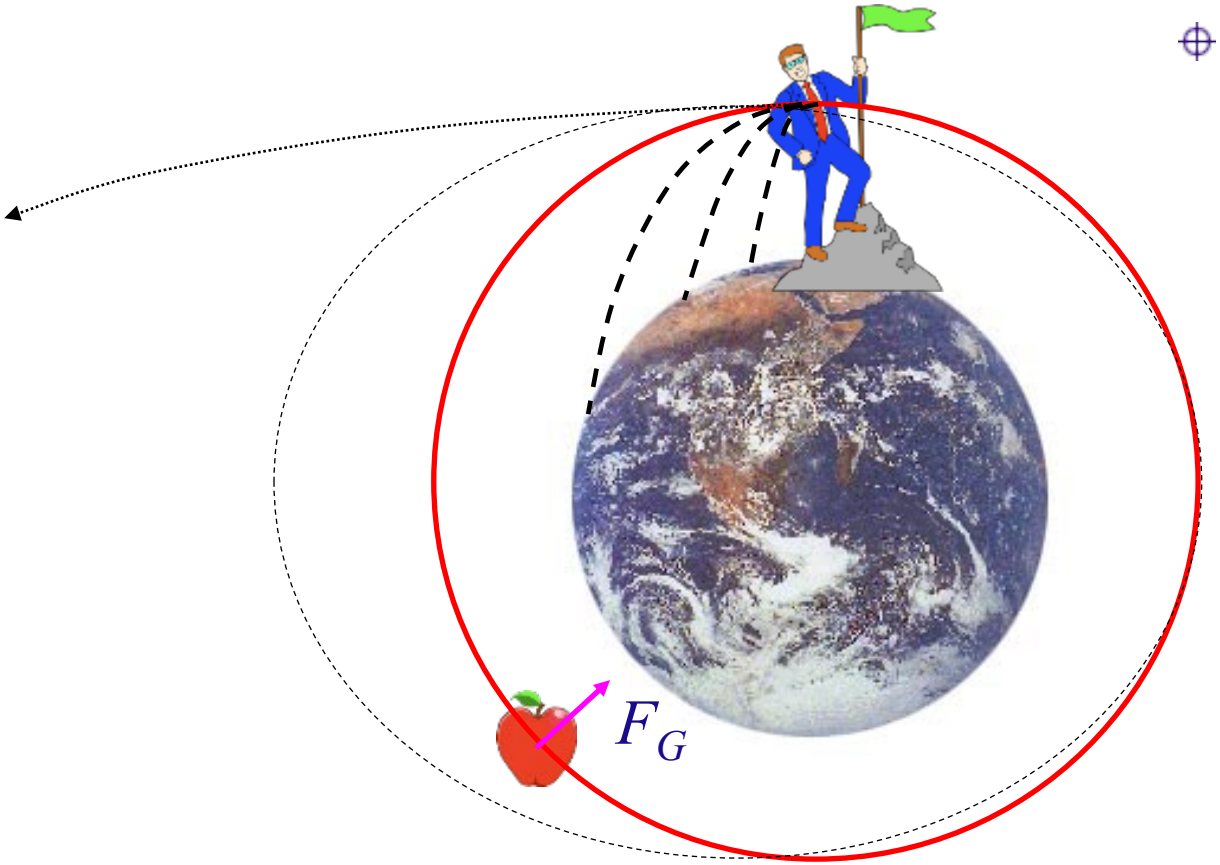
Orbits – G 5.7

Throw a ball horizontally…



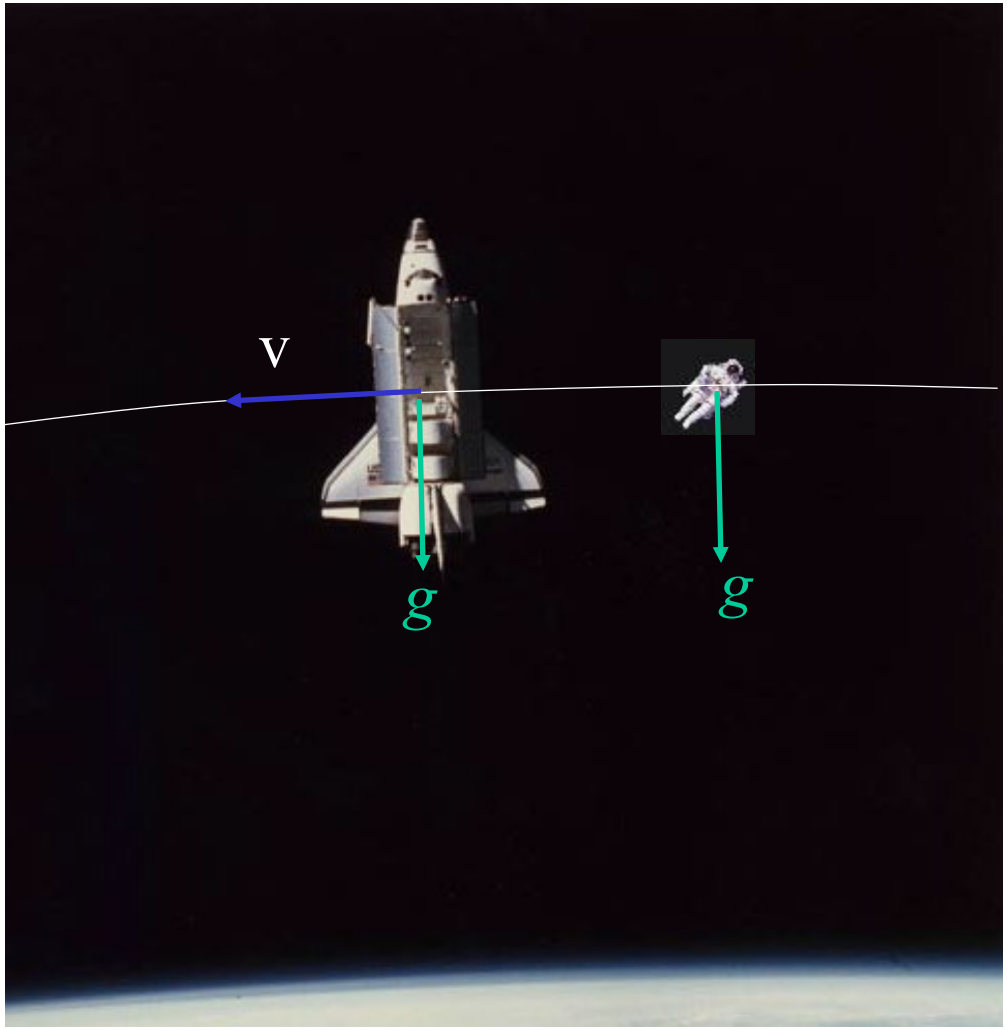
Orbits - G 5.7

Increase the speed of the throw…



- The apple is constantly falling towards the Earth, but Earth's surface curves away beneath it.

‘Weightlessness’ in orbit - G 5.7



$$a = g = \frac{Gm_E}{r^2} = \frac{v^2}{r}$$

Using data from previous slide $g = 8.7\ m/s^2$ – not a lot less than on the Earth’s surface!

Astronaut feels weightless because:

- He and the space shuttle both accelerate at the same rate.
- Both are in free-fall.
- There are no contact forces on the astronaut. Therefore his apparent weight is zero.

Conservation laws in physics

- *Conserved* quantities are very important in physics, because they allow us to predict the outcomes of complicated interactions without knowing all the details of the forces involved.

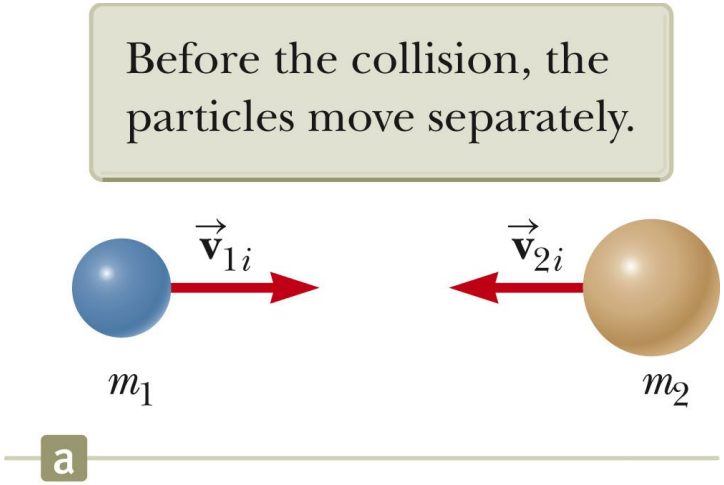
- When objects interact with one another, the details of the forces involved can be very complicated:
  - different forces can act at different stages of the interaction.
  - forces can vary in magnitude and/or direction during the interaction.

Conservation laws in physics

- Examples of conservation laws:
  - Conservation of *mass*
    - e.g. in a chemical reaction, the mass of the reactants must equal the mass of the products.
  - Conservation of *momentum*.
  - Conservation of *energy*.
  - Conservation of *angular momentum*.
    - (We examined this when we studied rotational motion.)

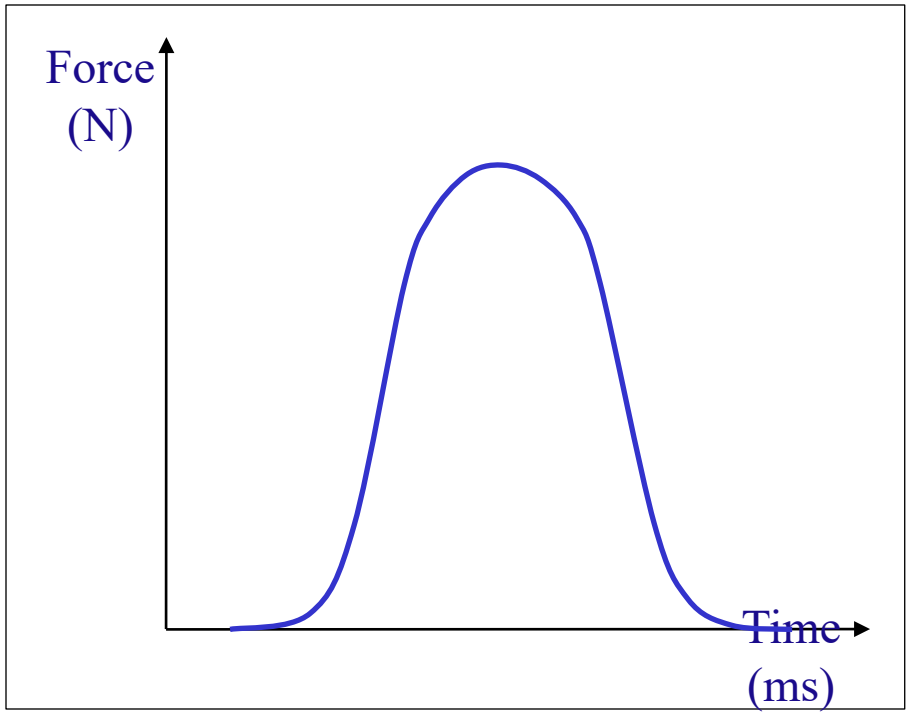
- Conservation laws provide us with an alternative way of looking at physical systems, not in terms of how things change, but in terms of what stays the same.
  - They also give us alternative and powerful new ways to solve problems, making some difficult problems much easier, and other seemingly-intractable problems solvable.

Collisions – G 7.3



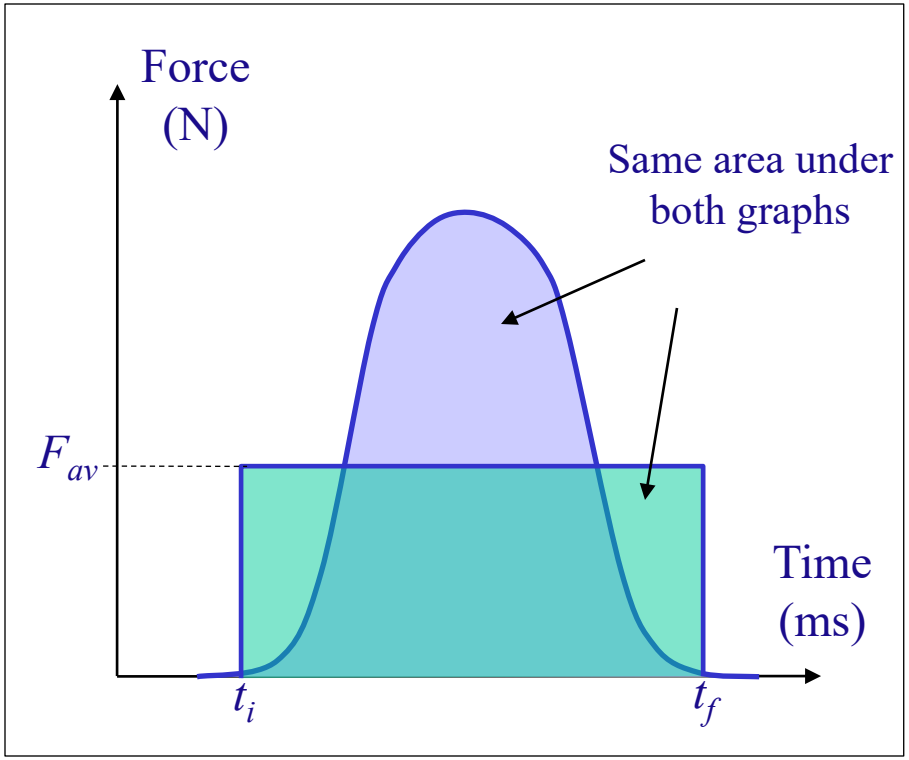
- A *collision* is a short-duration interaction between two objects.
- Typical collision times for macroscopic objects may be 1 ms – 100 ms (but there are exceptions).
- Forces between the objects involved form a Newton III action/reaction pair.
- The magnitude of the force on the object of interest usually varies with time.
  - e.g. think of the racquet as exerting a 'spring force' during the above collision.

Impulsive forces – G 7.3



- Impulsive force
  - A force exerted over a short period of time (Δt).
  - Large magnitude compared to any other forces simultaneously acting on the object of interest – e.g. maximum force of racquet on the ball is much greater than the weight force on the ball during the collision.

Impulse – G 7.3



- In general, impulse is a *vector* quantity that points in the same direction as the impulsive force.

$$\vec{J} = \vec{F}_{av}\Delta t$$

The *impulse* of a force is the area under the force-time graph.

$$J \equiv \int_{t_i}^{t_f} F dt$$

- Often, we don't know the precise shape of the force-time graph, but can think about the impulse in terms of the *average* force.
- Average force acts for the same time as the actual force, and has a value that gives the same impulse (same area under force-time graph).

$$J \equiv \int_{t_i}^{t_f} F dt \equiv F_{av}\Delta t$$

Linear momentum – G 7.1

The *linear momentum* of an object is:

$$\vec{p} \equiv m\vec{v}$$

momentum mass velocity

- If the word 'momentum' is used without specifying 'linear momentum' or 'angular momentum', we always mean linear momentum.
- Momentum is a *vector* which points in the same direction as the velocity vector.
- Units of momentum are kg m s^{-1}.
  - Since momentum depends on both mass and velocity, a slow-moving massive object can have the same momentum as a fast-moving object with low mass.

Impulse changes momentum – G 7.3

Suppose an impulsive force acts on an object (e.g. force of tennis racquet on ball).

$$\vec{J} = \vec{F}_{av}\Delta t$$

Impulsive force is the only force acting (it is the net force):

$$\vec{J} = m\vec{a}_{av}\Delta t$$

Average acceleration is rate of change of velocity:

$$\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t} \qquad \nabla\vec{v} = \vec{a}_{av}\Delta t$$

So $\vec{J} = m\Delta\vec{v} = m\left(\vec{v}_f - \vec{v}_i\right) = m\vec{v}_f - \mathrm{m}\vec{v}_i$

Impulse-momentum theorem: $\vec{J} = \vec{F}_{av}\Delta t = m\Delta\vec{v} = \Delta\vec{p}$

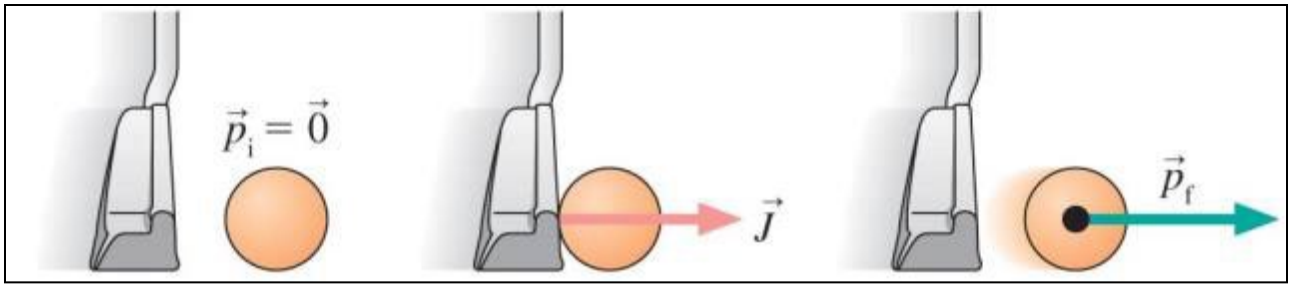
Impulse changes momentum – G 7.3

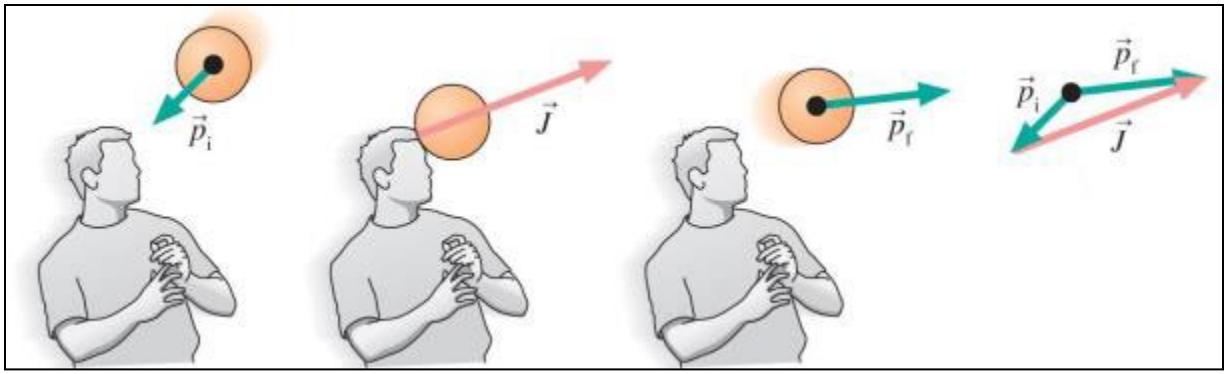
An impulsive force on an object causes the object's momentum to change.

Impulse-momentum theorem applies in vector form:

$$\vec{J} = \vec{F}_{av}\Delta t = m\Delta v = \Delta\vec{p}$$

$$\vec{J} = \Delta\vec{p} = \vec{p}_f - \vec{p}_i \qquad \vec{p}_f = \vec{p}_i + \vec{J}$$





PHYC10009 week3

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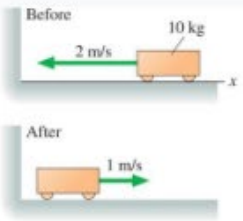
Cart collision



What impulse (in the x direction) does the wall deliver to the cart?

- A. −30 kg m/s.
- B. −20 kg m/s.
- C. −10 kg m/s.
- D. 10 kg m/s.
- E. 20 kg m/s.
- F. 30 kg m/s.

A

B

C

D

E

F

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Stopping a moving object - G 7.6

$$\vec{F}_{av}\Delta t = m\Delta\vec{v} = m(\vec{v}_f - \vec{v}_i)$$

$$(F_{av})_x = \frac{m(0-(v_i)_x)}{\Delta t} = \frac{m(0-(-v_i))}{\Delta t} = \frac{+mv_i}{\Delta t}$$

- Average force on the car (and passengers inside) depends on
  - Initial velocity.
  - How long the collision takes (Δt).
- To minimise the force for a given initial speed, increase the collision time.

Conservation of momentum - G 7.2

$$\vec{P} = M\vec{v} \qquad \vec{F} = M\vec{a}$$

$$\frac{d}{dt}\vec{P} = \frac{d}{dt}(M\vec{v}) = M\frac{d\vec{v}}{dt} = M\vec{a}$$

$$\vec{F}_{net\ ext} = \frac{d\vec{P}}{dt}$$

- Changes in the total momentum of a system are caused by the net external force.

- If the net external force on a system is zero, total momentum is conserved (i.e. has a constant value). Such a system is said to be *isolated*.

In summary…

Linear motion

$$F_{net\ ext} = \frac{dP}{dt}$$

Net force on a system determines its rate of change of linear momentum.

$$(F_{net\ ext})_{av} = \frac{\Delta P}{\Delta t}$$

$$\Delta P = F_{net}\Delta t$$

(Version of the above law for a finite time interval.)

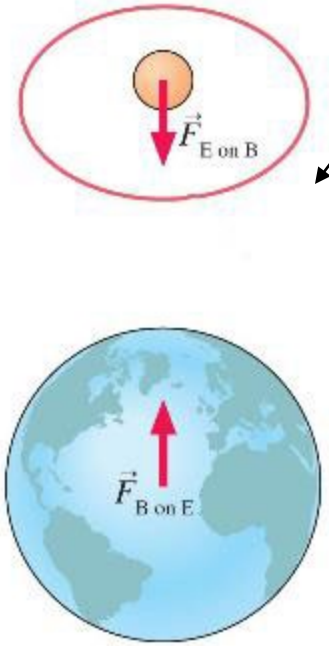
If $F_{net\ ext} = 0$, then $\Delta P = 0$.

Linear momentum of the system is conserved.

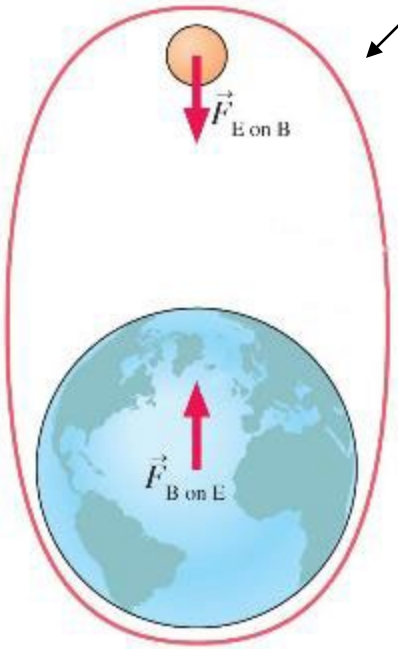
Final momentum = initial momentum.

Choosing the system…

- In order for conservation of momentum to apply, we must include enough objects in the system so that there are is no net *external* force acting on it.
  - Example: a ball falling from rest to the ground.



Gravitational force of Earth on ball is an external force on the ball. It delivers an impulse to the ball, causing its momentum to change. Momentum is not conserved.



The two gravitational forces are now *internal* forces. The *system* of Earth and ball is isolated, and total momentum is conserved. (It actually remains zero at all times. The ball gains a downwards momentum, and the Earth gains an equal and opposite upwards momentum.)

Momentum and Forces.

$$F_{net\ ext} = \frac{dP}{dt}$$

Net force on a system determines its rate of change of linear momentum.

$$(F_{net\ ext})_{av} = \frac{\Delta P}{\Delta t}$$

$$\Delta P = F_{net} \Delta t$$

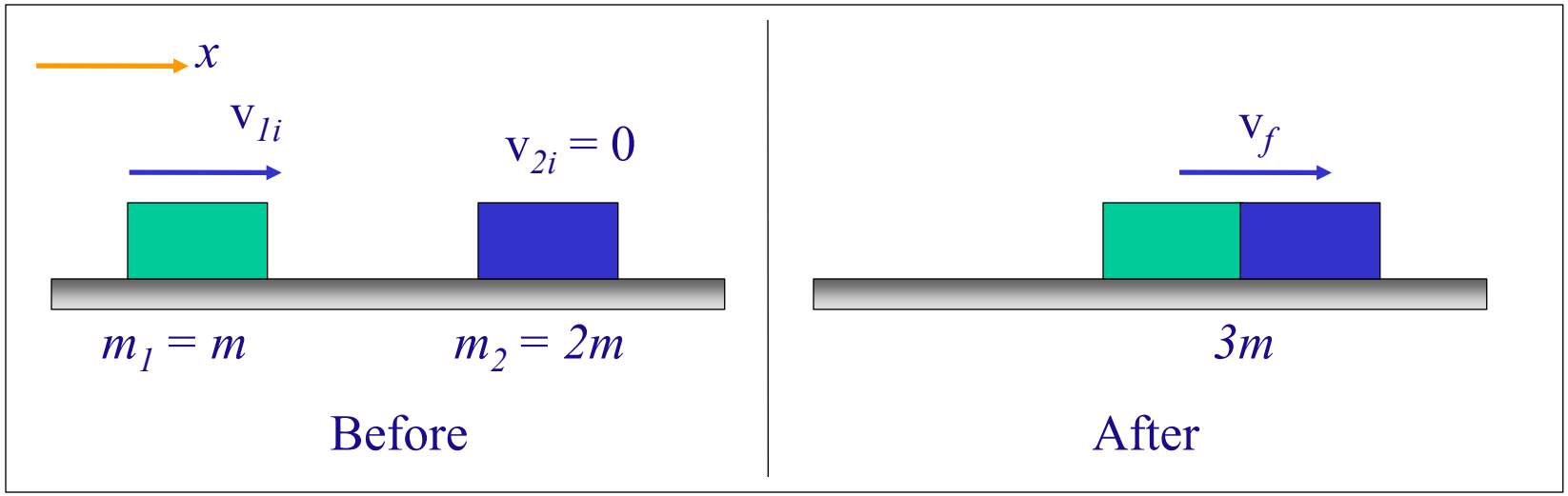
(Version of the above law for a finite time interval.)

If $F_{net\ ext} = 0$, then $\Delta P = 0$.

Linear momentum of the system is conserved.

Final momentum = initial momentum.

Example: air track collision – inelastic



Moving cart hits stationary cart, and they stick together. Final speed?

Net external force on system of two carts (weight + normal force) is zero, so momentum is conserved.

$$P_i = m_1 v_{1i} + m_2 v_{2i} = m v_{1i}$$

$$P_f = (m_1 + m_2) v_f = 3 m v_f$$

$$P_i = P_f$$

$$m v_{1i} = 3 m v_f$$

$$v_f = v_{1i} / 3$$

D

Mj-6 (Airtrack)